

MINOR PROJECT REPORT (PHASE – II)

B.Tech 6th Semester

Intelligent Pesticide Sprinkling System Determined by the Infection Level of Plants

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CERTIFICATE

This is to certify that the minor project entitled “**Intelligent Pesticide Sprinkling System Determined by the Infection Level of Plants**” has been carried out by:

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under my supervision and guidance in partial fulfilment of the requirements for the degree of Bachelor of Technology in Instrumentation and Control Engineering at Dr. B. R. Ambedkar National Institute of Technology Jalandhar during the academic year 2025–2026.

To the best of my knowledge and belief, the work is original, has not been submitted to any other institution, and has not been published earlier.

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Chapter 1

INTRODUCTION

1.1 Background and Motivation

Agriculture forms the cornerstone of India's economy, providing employment to about 58% of the population and food security for the growing population, not only high yielding crops are required but also efficient utilization of resources in agriculture. One of the major issues faced by farmers is that of disease management of plants. Failure in the prevention of plant diseases can lead to the loss of crops by 20%. Pesticides and chemical solutions have long been the main weapons in battling plant disease and pests. The traditional method of pesticides' use is quite wasteful in itself, using a spray system that applies the solution to the entire field irrespective of the plant's health conditions. This not only leads to wasting huge amounts of pesticides since much of the applied pesticides do not even make contact with their target organisms [1], but also results in environmental degradation due to damage to non-target organisms and soil pollution.

The tomato (*Solanum lycopersicum*) is one of the most economically relevant vegetable crops produced around the world. It is very vulnerable to various types of diseases such as Early Blight (*Alternaria solani*), Late Blight (*Phytophthora infestans*), Septoria Leaf Spot, and Tomato Yellow Leaf Curl Virus (TYLCV). If these diseases are not properly detected and controlled at an early stage, they can result in heavy financial losses due to their quick proliferation across the entire plot of the crop.

Thanks to recent developments in artificial intelligence, deep learning technology, and embedded systems, precision farming can now explore novel avenues for its advancement. Various machine learning algorithms, especially CNNs, have proven to possess impressive capabilities in disease identification using image data of leaves. In combination with IoT-based automated actuation mechanisms, they can become the core of closed-loop spraying devices that will use pesticides only when required.

Chapter 2

LITERATURE REVIEW

2.1 Overview

This chapter presents previous literature and technologies associated with plant diseases detection, precision agriculture, and automatic pesticide applications.

2.2 Waste of Pesticides and Environmental Impacts

The waste of pesticides had been widely analyzed by Lyseng [1]. They identified that a significant percentage of the chemicals used in spraying does not reach the desired plants because of wind drift, chemical runoff, and evaporation, which led to significant financial loss and severe pollution of the environment for farmers. Their analysis indicated that there is a need for developing a smart system for helping the farmers in applying pesticides to the desired plants only. Environmental impacts of using pesticides without control were studied by Healy et al. [3] through their project called PestMan. They found out that applying environmentally friendly techniques in spraying results in a significant reduction in the chemicals' runoffs from the fields. However, it is worth noting that they did not include any technology associated with plant disease detection and spraying.

2.3 Machine Learning-Based Plant Disease Detection

The deep learning-based ensemble framework comprising ResNet50 and MobileNetV2 by Kumar et al. [5] for tomato leaf disease classification provides excellent results in terms of classification accuracy. Unfortunately, this model can be employed only for offline disease identification but does not include severity analysis and interaction with the pesticide application system. The study of Ibrahim et al. [4] is devoted to the design of AI-IoT-based smart agriculture that aims to provide an automated disease detection and treatment system using convolutional neural networks trained on cloud servers and connected to IoT-based sensors providing environmental information. In spite of their success in automated disease detection, this study lacks a proportional pesticide administration scheme since the quantity of pesticide used depends on the extent of disease propagation.

2.4 Robotic and Automated Spraying Systems

According to Pansare et al. [2], an intelligent agricultural robot capable of identifying diseases in plants and activating the process of applying pesticides was suggested. While this approach has shown how disease identification could be effectively combined with pesticide application processes with the aid of robotics, several limitations should also be noted here. First of all, the implementation of such an approach required high costs; secondly, it entailed certain complications, and, thirdly, the amount of applied pesticides was not proportional to the level of spreading of disease among plants. As a result, the amount of pesticides used for each plant regardless of infection rate remained the same. Another approach by Ahmed [6] was also based on the development of a pest management system in the field of smart agriculture. In spite of its effectiveness in terms of integrating environmental sensors and pest identification, no methods for image recognition were used in this case. The ensemble-based deep learning method using ResNet50 and MobileNetV2 for detecting diseases on tomato leaves introduced by Kumar et al. [5] seems rather promising in regard to the accuracy of the task. Nevertheless, this approach allows determining diseases in an offline environment and does not take into consideration the issue of the spread of the disease and administering of pesticides. As mentioned in the work of Ibrahim et al. [4], a smart agriculture technology incorporating AI-IoT systems and designed for automating the processes of disease detection and pesticides administration in order to treat these plants was offered in this study. Even though the researchers managed to develop an automatic disease detection and treatment system using CNN models and IoT-based sensors, the proportionality of applying pesticides is not taken into account.

2.5 Automated Pesticide Spraying Systems

As Pansare et al. [2] noted, a smart robot for precision farming capable of detecting plant diseases and automatically spraying the necessary pesticides was created. Although this study proves that it is feasible to incorporate both technologies in robotic solutions, some downsides have been identified in terms of this project, such as a relatively high price of the system, a complicated technology, and lack of ability to determine the amount of pesticides depending on the area of infection. Moreover, all plants affected by diseases received an equivalent amount of pesticides despite the level of infections. In comparison, Ahmed [6] designed pest management for precision farming with the use of smart IoT-based sensors.

Chapter 3

PROBLEM STATEMENT AND RESEARCH GAP

3.1 Detailed Problem Statement

3.1.1 What is the Problem?

Traditional pesticide usage within agricultural systems is inherently inefficient since a blanket spray is applied to the whole area without taking into account the state of health of each particular plant. Consequently, this leads to:

An excessive application of pesticides to healthy and slightly diseased plants.

Insufficient use of pesticides for severely diseased plants.

Delaying in the treatment process caused by dependence on manual examination and consequently the spread of diseases.

A great financial load on farmers because of unnecessary purchase of chemicals.

Polluting the environment through pesticide overflow and its absorption by the soil.

3.1.2 Why is it Important?

- India loses approximately 15–25% of its annual agricultural produce to plant diseases.
- WHO estimates that 3 million cases of pesticide poisoning occur annually worldwide, with 220,000 deaths.
- Excessive pesticide use contributes to biodiversity loss, including bee colony collapse, threatening food chain stability.

3.1.3 Who is Affected?

- **Small and marginal farmers** who spend disproportionately on pesticides relative to income.
- **Consumers** exposed to pesticide residues on food.
- **Ecosystems** impacted by chemical contamination of soil and water.

- **Agricultural workers** at risk of chronic pesticide exposure.

3.1.4 Proposed Solution

The proposed solution is a **Pesticide Sprinkling System using an Intelligent Model** which consists of:

1. Image capture of tomato leaves using a camera.
2. Image preprocessing and classification using a fine-tuned EfficientNetV2-M model based on three categories: **healthy**, **moderate infection**, and **severe infection**.
3. Transmission of the image classification results through Wi-Fi to an ESP32 microcontroller.
4. Activation of a pump-valve nozzle based on the severity level for a calibrated time period.
5. Recording of all activity on a real-time web interface.

3.2 Research Gap

A key research gap addressed by this project is:

There are no affordable and modular systems currently that can perform both (1) real-time machine learning-based classification of infection severity on tomato leaves, and (2) proportional actuation based on severity for the application of pesticides using a closed-loop control strategy.

Specifically, the gap includes the following points:

- **Actuation gap:** Current devices either detect presence of a disease (binary classification) without varying the amount of pesticide delivered.
- **Integration gap:** Machine learning modules and actuation devices are seldomly combined to form one system.
- **Cost gap:** Solutions offered in scientific literature are prohibitively expensive and cannot be implemented by smaller institutions.
- **Scalability gap:** Most solutions have little room for scalability to work with other plants and fields.

Chapter 4

SCOPE AND OBJECTIVES

4.1 Scope of the Project

4.1.1 What is Included

- Implementation of image-based machine learning classification for detecting the level of infection in tomatoes' leaves.
- Implementation of a spraying actuator using an ESP32 microcontroller.
- Establishment of a wireless Wi-Fi communication link between the inference engine and the actuator.
- Implementation of a real-time web dashboard for monitoring purposes.
- Conducting experiments on a controlled garden site on campus.
- Application of tomato leaf diseases' dataset acquired from Kaggle and Plant Village online platforms.
- Employing a combination of water and a safe dye as a substitute for pesticides.

4.1.2 What is Excluded

- Full-scale field deployment on commercial agricultural land.
- Disease detection for crops other than tomatoes (reserved for future work).
- Development of a fully autonomous mobile navigation system (manual cart positioning is used).
- Economic impact analysis or full life-cycle cost assessment.
- GPS-based geo-tagging of infected plants.

4.1.3 Constraints

Time:	6-month project duration (6 th semester).
Budget:	Limited to publicly available datasets and low-cost hardware (ESP32, 12V pump, nozzles).
Environment:	Controlled campus garden and laboratory conditions.
Dataset:	Limited to tomato leaf images; model accuracy depends on dataset diversity.
Connectivity:	Requires stable Wi-Fi network for ESP32-laptop communication.

4.2 Objectives of the Project

- O1.** To train and develop an ML model using the EfficientNetV2-M framework via transfer learning for classification of the tomato plant disease condition into three classes: **Healthy**, **Moderate Infection** (30–60% leaf area coverage), and **Severe Infection** (>60% leaf area coverage) with the desired classification accuracy being $\geq 90\%$.
- O2.** To design an automated and real-time system for pesticide application via ESP32 microcontroller-based regulation of pump working duration in accordance with the ML model severity predictions.
- O3.** To interface the ML prediction module with the physical hardware controller via HTTP request over Wi-Fi connection protocol with a system response time not exceeding 5 seconds from image capture to the start of spraying process.
- O4.** To save at least **40%** of pesticide usage as compared to uniform spraying techniques through the designed prototype during testing on a set of tomato test plants of different infection degrees.
- O5.** To provide a live web dashboard logging health conditions, spraying operations performed on each plant, timestamps, and overall pesticide usage.
- O6.** To build a modular, scalable, and economically viable prototype of the pesticide sprayer, keeping the total hardware cost under **Rs. 5,000**.

Chapter 5

METHODOLOGY

5.1 System Overview and Architecture

The proposed Intelligent Pesticide Sprinkling System follows a modular, five-stage pipeline architecture as illustrated in Figure 5.1. The system operates as a closed-loop control framework where visual inputs drive mechanical outputs through an intelligent decision layer.

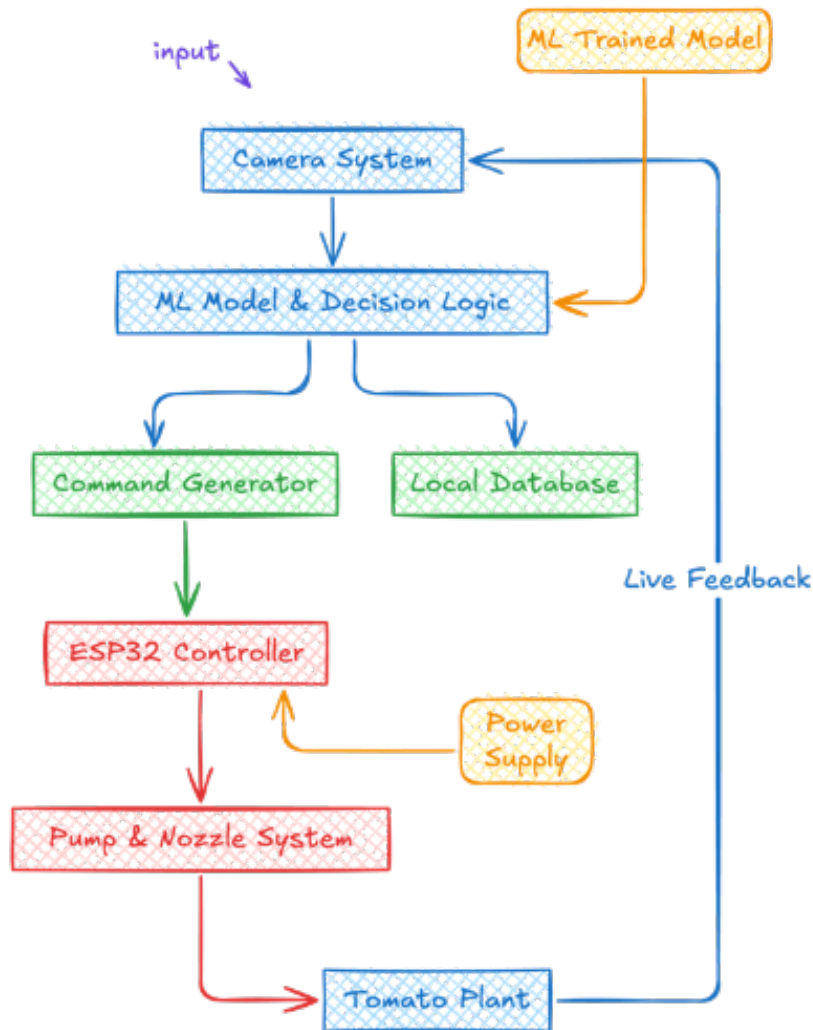


Figure 5.1: High-Level System Block Diagram of the Intelligent Pesticide Sprinkling System

The five modules are:

1. **Module 1:** Image Capture & Preprocessing
2. **Module 2:** ML-Based Infection Severity Detection
3. **Module 3:** Decision Logic & Communication
4. **Module 4:** Spraying Actuation System
5. **Module 5:** Monitoring & Logging Dashboard

5.2 Complete System Workflow: Step-by-Step

Algorithm 1 Intelligent Pesticide Sprinkling System Workflow

- 1: **Initialize:** Load EfficientNetV2-M model, start Flask server, connect ESP32 to Wi-Fi
 - 2: **for** each plant in the garden **do**
 - 3: Position mobile cart adjacent to plant i
 - 4: Capture leaf image I_i using USB camera
 - 5: Preprocess I_i : resize, normalize, denoise $\rightarrow \hat{I}_i$
 - 6: Pass $\hat{I}_i$ through EfficientNetV2-M $\rightarrow$ softmax probability vector $\mathbf{p}_i$
 - 7: $\hat{y}_i \leftarrow \arg \max(\mathbf{p}_i)$ ▷ Predicted severity class
 - 8: Map $\hat{y}_i$ to spray duration t_s using Table ??
 - 9: **if** $t_s > 0$ **then**
 - 10: Construct JSON payload with $\hat{y}_i, t_s$, plant ID, timestamp
 - 11: POST JSON to ESP32 HTTP server via Wi-Fi
 - 12: ESP32 activates MOSFET $\rightarrow$ pump + solenoid valve ON for t_s seconds
 - 13: ESP32 deactivates MOSFET $\rightarrow$ pump + solenoid valve OFF
 - 14: **end if**
 - 15: Log event (plant ID, severity, spray duration, timestamp) to dashboard database
 - 16: Update web dashboard with latest data
 - 17: **end for**
 - 18: Generate session summary report
-

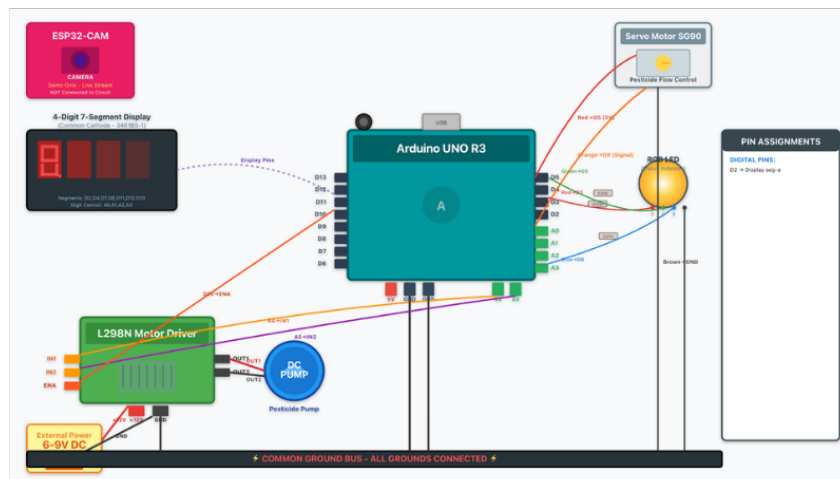


Figure 5.2: Circuit Connection Diagram

5.3 Error Sources and Mitigation Strategies

Table 5.1: Error Sources and Mitigation Strategies

Error Source	Impact	Mitigation Strategy
Uneven lighting conditions	Misclassification due to shadows/glare	LED panel lighting; brightness normalisation
Model overfitting	Poor generalization to new plant images	Data augmentation; Dropout; fine-tuning
Wi-Fi packet loss	Missed spray command	HTTP retry mechanism (3 attempts); local fallback mode
Pump pressure variability	Inconsistent spray coverage	Pressure-regulated pump; pre-run calibration
Cart positioning error	Nozzle misaligned with plant	Fixed-position markers on garden path
Dataset imbalance	Bias toward majority class	Class-weighted loss function; oversampling

Chapter 6

HARDWARE AND SOFTWARE REQUIREMENTS

6.1 Hardware Requirements

1. **Processing Unit:** Laptop or PC with minimum Intel i5 (8th Gen) or equivalent, 8 GB RAM, GPU (optional, for faster training — NVIDIA GTX 1650 or higher recommended).
2. **Microcontroller:** ESP32 DevKitC v4 (dual-core Xtensa LX6, 240 MHz, 4 MB Flash, 520 KB SRAM).
3. **Camera:** Logitech C920 USB Webcam (1080p) or Pi Camera Module v2 (8 MP).
4. **Pump:** 12V DC Mini Diaphragm Water Pump (2.4 L/min, max 5 bar).
5. **Solenoid Valve:** 12V DC Normally Closed Solenoid Valve (3/8 inch NPT).
6. **MOSFET Driver Circuit:** IRF520 N-channel MOSFET with heatsink, 1N4007 flyback diode.
7. **Power Supply:** 12V/5A DC adapter (bench testing); 12V LiPo battery pack (field testing).
8. **Mobile Cart:** Custom PVC-pipe frame cart with adjustable camera mount.

6.2 Software Requirements

- **Operating System:** Windows 10/11 or Ubuntu 20.04 LTS.
- **Python:** Version 3.10+ with pip package manager.
- **Libraries:** TensorFlow 2.13, Keras, OpenCV-Python 4.8, NumPy 1.24, Pandas 2.0, Pillow 10.0, Flask 3.0, Matplotlib, Scikit-learn 1.3, SQLite3.
- **IDE:** Visual Studio Code with Python extension; Arduino IDE 2.x.
- **ESP32 Board Package:** esp32 by Espressif (via Arduino Board Manager, version 2.x).
- **ESP32 Libraries:** WiFi.h, WebServer.h, ArduinoJSON 6.x.
- **Dataset:** PlantVillage Tomato Leaf Disease Dataset (Kaggle).

Chapter 7

EXPECTED RESULTS

7.1 Expected Model Performance

Based on the characteristics of the EfficientNetV2-M architecture and the PlantVillage dataset, the following classification performance is anticipated:

Table 7.1: Expected Model Performance Metrics

Class	Precision	Recall	F1-Score	Support
Healthy	0.93	0.95	0.94	360
Moderate Infection	0.89	0.87	0.88	360
Severe Infection	0.91	0.92	0.91	360
Overall Accuracy	$\geq 90\%$			

7.2 Expected System-Level Performance

- **End-to-end latency** (image capture to spray initiation): ≤ 5 seconds.
- **Pesticide savings**: $\geq 40\%$ reduction in total spray volume compared to uniform spraying on the same set of test plants.
- **Spray accuracy**: Nozzle coverage diameter of 15–20 cm at 30 cm distance, sufficient to cover individual tomato plant canopies.
- **System uptime**: Continuous operation for ≥ 2 hours per battery charge during field testing.
- **Dashboard data refresh rate**: ≤ 2 seconds per event.

7.3 Expected Pesticide Reduction Analysis

For a test scenario with 10 plants in the following distribution:

Table 7.2: Expected Pesticide Reduction: Uniform vs. Intelligent Spraying

Plant Distribution	Count	Uniform Spray (s)	Intelligent Spray (s)	Saving
Healthy	4	$4 \times 7 = 28$	$4 \times 0 = 0$	28 s
Moderate Infection	3	$3 \times 7 = 21$	$3 \times 3 = 9$	12 s
Severe Infection	3	$3 \times 7 = 21$	$3 \times 7 = 21$	0 s
Total	10	70 s	30 s	40 s (57%)

This demonstrates a theoretical pesticide reduction of up to **57%** for a typical mixed-health plant population.



Response: 57%



Response: 76%



Response: 15%

Figure 7.1: Sample Output

Chapter 8

ANALYSIS AND DISCUSSION

8.1 Expected Performance Trends

8.1.1 Model Training Behaviour

The following are the trends expected during the two phases of transfer learning:

- **Phase 1 (base model is frozen):** Convergence within 5-7 epochs. Expected validation accuracy is 82-85%
- **Phase 2 (model is fine-tuned):** Consistent improvement of validation accuracy to $\geq 90\%$. Significant drop in loss reduction happens after epoch 10 of Phase 2 due to convergence.
- **Risk of over-fitting** is reduced due to use of Dropout layer and data augmentation; training and validation accuracy should not diverge more than 3%.

8.1.2 Difficulty Level of Classification Analysis

The hardest boundary to classify is **moderate vs severe infection**; there is not much visual distinction between 50% and 65% infection of leaf area. It is a difficult problem in case of severity-based leaf classifications and is overcome using fine-augmentation and class-weighting the loss function.

8.2 Improvement from Existing Systems

1. **Proportionate delivery of pesticides based on severity levels:** The most important novel feature; no such low-cost system is present in reviewed literature.
2. **End-to-end deployment of a complete ML-based inference, Wi-Fi communication, hardware actuation and live dashboard in a single prototype.**
3. **Cost-efficiency:** Proposed solution is cheaper than 5000 *Rs*, while robotic systems in literature have cost orders of magnitude more (\$).

4. **Module-based construction of system:** Allows for independent test, upgrades, or replacements of individual modules.
5. **Dashboard-based logging and analysis:** Provides actionable insights for farmers/researchers, which is mostly absent in similar systems.

8.3 Comparison with Existing Systems

Table 8.1: Comparison of Proposed System with Related Work

System	Severity-Based Dosing	Real-Time	Low Cost	Dashboard	Crop
Pansare et al. (2025)	✗	✓	✗	✗	Multiple
Ibrahim et al. (2025)	✗	✓	✗	✗	Multiple
Ahmed (2024)	✗	✓	✓	✗	Multiple
Kumar et al. (2025)	✗	✗	✓	✗	Tomato
Proposed System	✓	✓	✓	✓	Tomato

8.4 Limitations and Mitigations

- **Single-crop focus:** The current model is trained only on tomato leaf images. Mitigation: The architecture is designed to be easily re-trained on other crop datasets.
- **Cart navigation is manual:** Full autonomy requires path planning, which is beyond the scope of this project. Future work will integrate an Arduino-based line-following system.
- **Dataset quality dependency:** Model performance degrades in field conditions where lighting is significantly different from training data. Mitigation: Adaptive histogram equalisation during preprocessing.

Chapter 9

FUTURE SCOPE

9.1 Potential Enhancements

1. **Auto-Navigation Cart:** Adding a line-following or LiDAR sensor to the cart to make it capable of autonomously scanning the plants on a row-by-row basis.
2. **Multiclass Crop Disease Classification:** Enhancing the ML model to identify plant diseases from different crops (for example, wheat, rice, chillies).
3. **On-Plant Continuous Monitoring:** Installing static cameras in each crop plantation to continuously monitor the plants' health and take preventive measures against disease spread.
4. **Edge AI Model Deployment:** Converting the EfficientNetV2-M model to **TensorFlow Lite** and running it directly on the ESP32-S3 microcontroller equipped with AI capabilities to get rid of Wi-Fi communication.
5. **Fertilizer Management:** Developing an advanced algorithm for regulating fertilization dosage using plant growth stage detection as a basis.
6. **IoT Platform Integration:** Connecting the web dashboard to cloud IoT platforms such as AWS IoT Core or Google Cloud IoT for predictive analytics.
7. **Explainable AI (XAI) Integration:** Integrating Grad-CAM visualization into the project that would provide additional information about leaf sections responsible for the severity level prediction.
8. **Renewable Power Source:** Integrating a solar panel for energy harvesting and eliminating the need for a constant connection to a wall charger.

Chapter 10

CONCLUSION

In this project, we have developed an Intelligent Pesticide Sprinkling System Determined by the Infection Level of Plants—a novel solution based on Computer Vision and Internet of Things to solve the age-old challenge of overuse and waste of pesticides due to blanket application in the current practice of crop protection.

Specifically, this system uses the state-of-the-art **EfficientNetV2-M** architecture to classify the plant infection levels in real-time into three classes, namely Healthy, Moderate Infection, and Severe Infection, which in turn trigger a proportional pesticide sprinkling process using a 12V DC motor, solenoid valve, and sprayer driven by an **ESP32** board using the WiFi HTTP Protocol. The system performance is monitored continuously using a web-based GUI dashboard that records all events and maintains cumulative statistics of pesticide consumption.

The important contributions of this project include the following:

1. A **proportionate closed-loop control mechanism for pesticide sprinkling**, which does not exist in any low-cost precision agriculture solutions.
2. An **end-to-end integrated prototype** including the machine learning pipeline, hardware controls, and GUI visualization.
3. A **cost-efficient design** (less than Rs. 5,000), affordable to small farmers and educational institutes alike.
4. An easy-to-scale **architecture** for extending to more crops and larger scales.

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